

SQL Mini Project

Online Food Delivery Management System

Project Overview

Nowadays, online food delivery applications are widely used by customers to order food from nearby restaurants. The company wants to maintain all customer orders, restaurant details, delivery information, payments, and order tracking using a database system.

This project helps students understand how SQL is used in real-time applications like food delivery platforms.

Examples:

- Swiggy
 - Zomato
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Real-Time Scenario

A food delivery company needs a database system to manage:

- Customer details
- Restaurant information
- Food items
- Orders placed by customers
- Delivery partner details
- Payment records
- Order status tracking

The system should help the company generate reports, analyze sales, rank restaurants, and track customer orders efficiently.

Modules in the Project

1. Customer Management

Stores:

- Customer ID
 - Customer Name
 - Mobile Number
 - Address
 - City
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2. Restaurant Management

Stores:

- Restaurant ID
 - Restaurant Name
 - Location
 - Rating
 - Contact Number
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3. Food Items Management

Stores:

- Food ID
 - Food Name
 - Category
 - Price
 - Restaurant ID
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4. Order Management

Stores:

- Order ID
 - Customer ID
 - Food ID
 - Quantity
 - Total Amount
 - Order Date
 - Order Status
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5. Delivery Partner Management

Stores:

- Delivery Partner ID
 - Partner Name
 - Phone Number
 - Assigned Order
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6. Payment Management

Stores:

- Payment ID
 - Order ID
 - Payment Method
 - Payment Status
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Database Tables

Customers

Contains customer personal details.

Restaurants

Contains restaurant information.

Food_Items

Contains available food item details.

Orders

Contains customer order information.

Delivery_Partners

Contains delivery staff information.

Payments

Contains payment transaction details.

SQL Concepts Used in This Project

DDL Commands

Used for:

- Creating database
- Creating tables
- Adding constraints
- Altering tables
- Dropping tables

DML Commands

Used for:

- Inserting records
- Updating order status
- Deleting cancelled orders

DQL Commands

Used for:

- Viewing customer orders
- Finding top restaurants
- Displaying delivered orders
- Calculating total sales
- Filtering data
- Sorting reports

Joins Used

INNER JOIN

Used to display:

- Customer name with ordered food item

- Restaurant details with orders

LEFT JOIN

Used to find:

- Customers who did not place orders

RIGHT JOIN

Used to display:

- All restaurants including those without orders

SELF JOIN

Used for:

- Restaurants located in same city
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Views Used

Customer_Order_View

Displays:

- Customer name
- Restaurant name
- Food item
- Order status

Restaurant_Sales_View

Displays:

- Restaurant-wise total sales

Delivery_Status_View

Displays:

- Delivery partner details with order status
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Window Functions Used

ROW_NUMBER()

Used to generate order sequence.

RANK()

Used to rank restaurants based on sales.

DENSE_RANK()

Used for customer ranking based on number of orders.

LEAD()

Used to compare next order amount.

LAG()

Used to compare previous order amount.

TCL Commands Used

Used for:

- Order cancellation rollback
- Payment confirmation commit
- Savepoint creation during transactions

Commands:

- COMMIT
 - ROLLBACK
 - SAVEPOINT
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DCL Commands Used

Used for database security:

- GRANT order access to employee
- REVOKE update permission

Expected Outputs

The system should be able to:

- Manage customer food orders
 - Track delivery status
 - Generate restaurant sales reports
 - Identify top-selling restaurants
 - Find active customers
 - Monitor payment status
 - Rank restaurants based on revenue
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Real-Time Benefits

This database system helps food delivery companies to:

- Reduce manual work
 - Track orders efficiently
 - Improve customer service
 - Analyze sales performance
 - Maintain secure transactions
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Learning Outcome

After completing this project, students will understand:

- Real-time database implementation
 - Relationships between tables
 - SQL joins and analytical functions
 - Transaction handling
 - Data security using DCL
 - Report generation using views
 - Real-world SQL project development
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Conclusion

The Online Food Delivery Management System is a real-time SQL mini project designed for students to practice database concepts in a practical way. This project combines multiple SQL concepts including joins, views, window functions, DML, DDL, DCL, TCL, and report generation.

It helps students gain confidence in designing and managing real-world database systems used in modern applications.